Hugo Marquerie 01/11/2025

Lem convergencia uniforme imp ctp

Lema 1. *La convergencia uniforme implica la convergencia puntual.*

***Demostración*:** Sean X, Y espacios métricos y $\forall n \in \mathbb{N} : f_n : X \to Y$ tales que $f_n \overset{X}{\underset{n\to\infty}{\rightrightarrows}} f$

$$\implies \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n \geqslant N : \forall x \in X : d_Y\left(f_n(x), f(x)\right) < \varepsilon.$$

En particular, $\forall x \in X : \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n \geqslant N : d_Y\left(f_n(x), f(x)\right) < \varepsilon$, luego $\forall x \in X : f_n(x) \xrightarrow[n\to\infty]{} f(x)$. ■

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